Spacing

Design rules demand a minimum gap between adjacent shapes that project on an axis

The idea

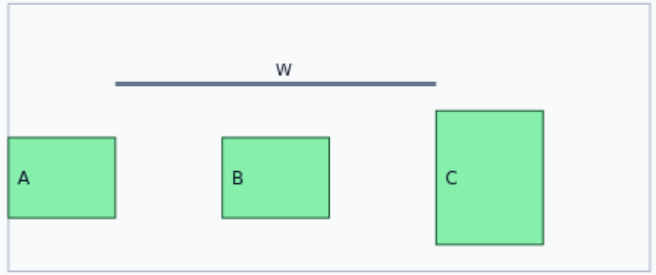
- Min_gap=1 on the toy
- Initial layout is legal; illegal if two cells overlap or gap < 1

Gap one

SPACING

Gap one

Step 1 / 5 · gap · module01-03-spacing-rules



Explanation

GOLDEN min_gap = 1.

- rule

```
gap=1
```

Gap one

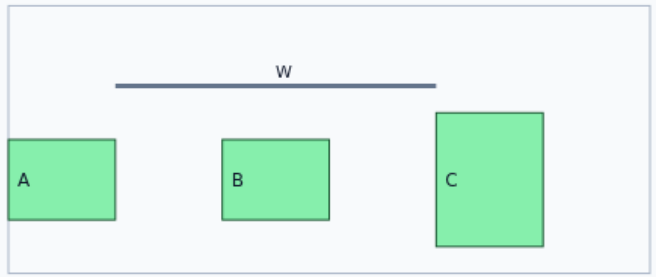
Legal init

SPACING

Legal init

Step 2 / 5 · legal · module01-03-spacing-rules

legal=true



Explanation

Starter layout is_legal → True.

- ok

```
True
```

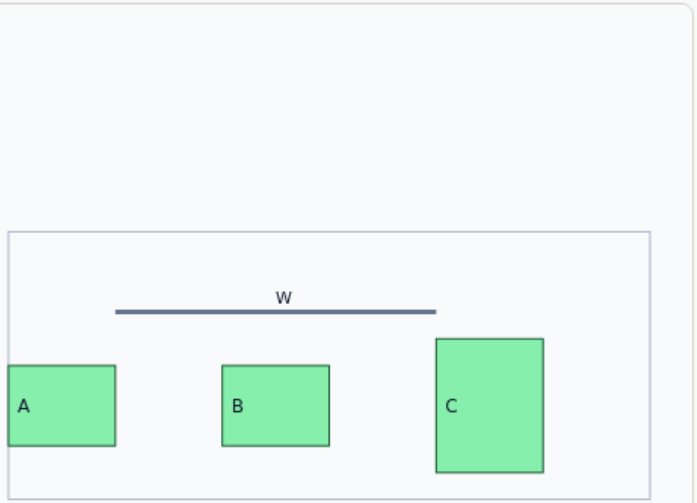
Legal init

Overlap fail

SPACING

Overlap fail

Step 3 / 5 · overlap · module01-03-spacing-rules



Explanation

Overlapping rects fail legality.

- reject

```
dx=dy=0
```

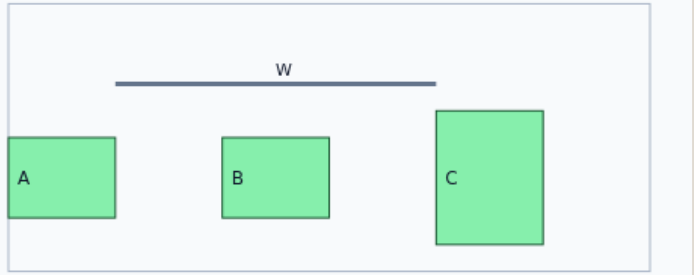
Overlap fail

Project

SPACING

Project

Step 4 / 5 · proj · module01-03-spacing-rules



Explanation

Spacing applies when orthogonal projections overlap.

- filter

```
Y overlap for X gap
```

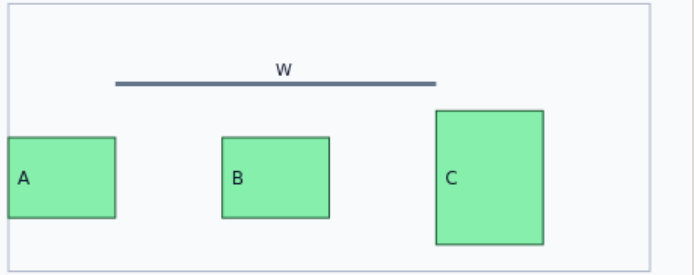
Project

Why

SPACING

Why

Step 5 / 5 · why · module01-03-spacing-rules



Explanation

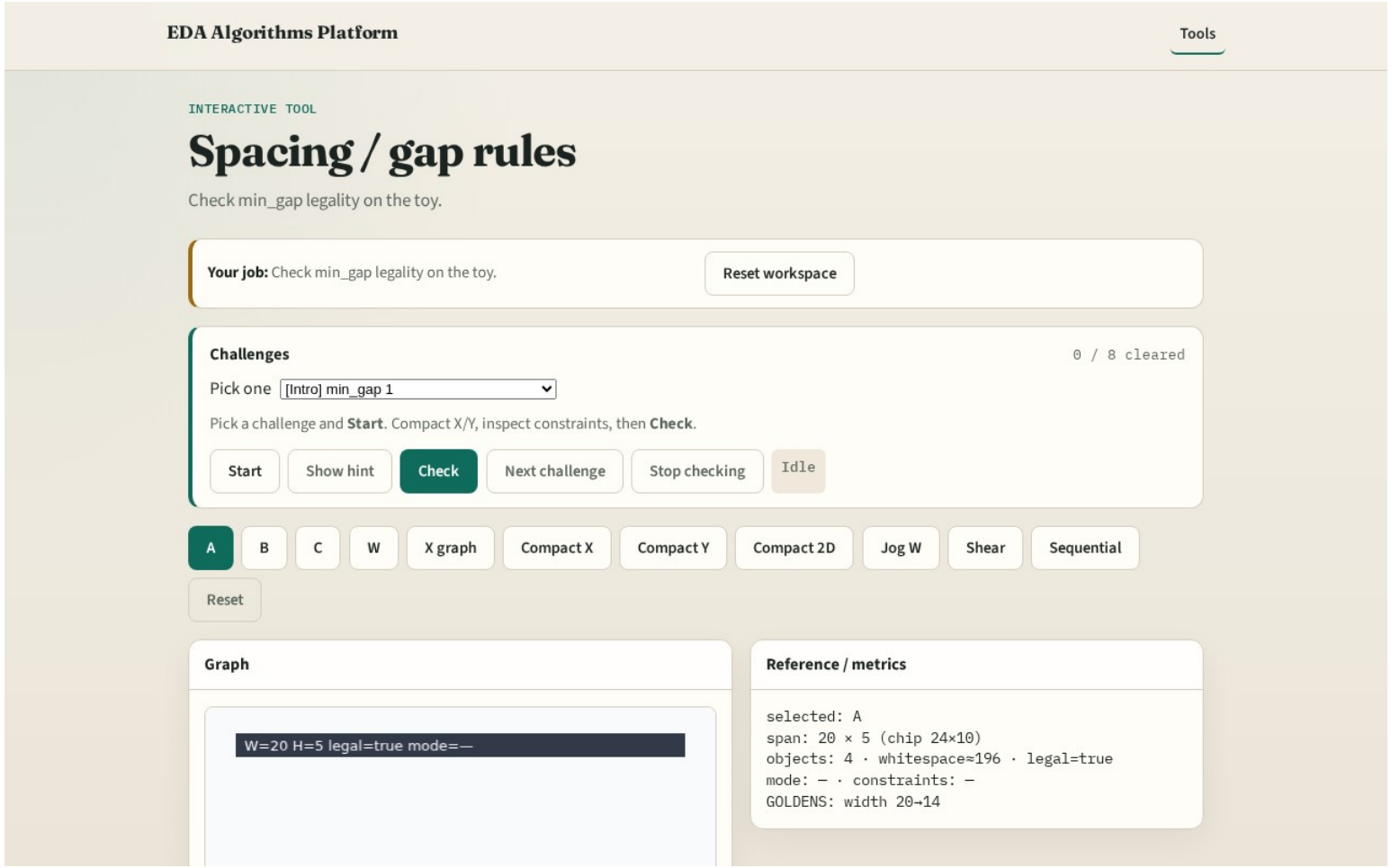
Constraints encode these gaps in the graph next.

- next

```
graph
```

Why

Browser lab track



Browser lab starter

Implement track

- Call is_legal(DATA); assert True
- Print min_gap

Pitfalls

- Applying gap without projection overlap
- Ignoring wires wrongly as cells

Your turn

- Next: the constraint graph